

A white Tesla Model S is parked on a grassy field. The car is positioned in the lower-left to center area of the frame. The background features a dense forest of green trees under a blue sky with some clouds. A semi-transparent teal and blue gradient overlay covers the top half of the image, serving as a background for the text.

Tesla Inc.

Company Valuation

A Detailed Walkthrough

Tesla Valuation - Sources

1	10 Year Treasury Yield	https://www.cnbc.com/quotes/?symbol=US10Y
2	Tesla's Beta, Share Price Shares Outstanding	https://finance.yahoo.com/quote/TSLA/ https://finance.yahoo.com/quote/tsla/key-statistics/
3	Tesla's Yield to Maturity	https://www.bondsupermart.com/main/bond-info/bond-factsheet/USU8810LAA18
4	Corporate Tax Rates	https://www2.deloitte.com/content/dam/Deloitte/global/Documents/Tax/dttl-tax-corporate-tax-rates.pdf
5	Tesla's Annual Reports (10k Filings)	https://ir.tesla.com/sec-filings?field_nir_sec_form_group_target_id%5B%5D=471&field_nir_sec_date_filed_value&items_per_page=50
6	Tesla's Sales Volumes	https://www.statista.com/statistics/502208/tesla-quarterly-vehicle-deliveries/
7	Model 3 Sales' Volume Estimate	https://insideevs.com/news/391128/tesla-model-3-cumulative-sales-best/#:~:text=In%202019%2C%20Tesla%20achieved%20an,of%20all%20cars%20since%202008
8	Tesla Vehicle Production Deliveries Estimate 2020	https://ir.tesla.com/news-releases/news-release-details/tesla-q1-2020-vehicle-production-deliveries
9	Annual Vehicle Deliveries - Comparables	https://www.statista.com/statistics/225326/amount-of-cars-sold-by-general-motors-worldwide/
10	Average Prices for Tesla	Solar Views Price Table Tesla Semi Roadster 2 https://insideevs.com/news/317916/comparing-a-base-tesla-model-s-to-a-fully-loaded-model-s-performance-plus-version-whats-the-price-d/ https://www.reuters.com/article/us-pepsico-tesla-orders-idUSKBN1E61FB
11	Annual Reports: BMW; Daimler; FCA; Ferrari; Ford; GM; Jaguar; Paccar; VW	Companies' websites

Inputs



Tesla Valuation – Drivers

365  Careers

Drivers

Data Validation >> List >> Values: 1,2,3

Selected case	2
Company	Tesla
Currency	USD
Domestic country	United States
10-year treasury yield (Sep 18 2020)	0.70%
Market risk premium US	5%
Company beta (Dec 15 2020)	2.15
Tesla share price (Dec 15 2020)	639.83
Bond Yield (Sep 18 2020)	4.32%
Corporate tax rate	21%
Expected inflation	2%

A “Drivers” sheet helps us organize some inputs we will need when working on the valuation model.

U.S. 10 Year Treasury (US10Y:U.S.)

Real Time Quote | Exchange

Yield | 4:59:21 PM EDT

0.697% +0.015



Bond Calculator

Settlement Date	Nominal Value	Buy At (USD)	Calculate
19 Sep 2020	2000	104.18	

→ You Pay

Estimated Total Payment Amount ⓘ	Indicative Accrued Interest ⓘ
USD 2,094.20	USD 10.60

→ You Get

Estimated Annual Coupon	Amount Received at Maturity	Estimated Yield to Maturity ⓘ	Next Call Date:	Next Call Yield:
USD 106.00	USD 2,000.00	4.324 %	15 Aug 2021	3.589

Tesla, Inc. (TSLA)

NasdaqGS - NasdaqGS Real Time Price. Currency in USD

 Add to watchlist

639.83 +29.84 (+4.89%) **644.20** +4.37 (0.68%)

At close: December 14 4:00PM EST

Before hours: 9:09AM EST

[Summary](#) [Company Outlook](#)  [Chart](#) [Conversations](#) [Statistics](#) [Historical Data](#)

Previous Close	609.99	Market Cap	606.496B
Open	619.00	Beta (5Y Monthly)	2.15
Bid	647.07 x 1800	PE Ratio (TTM)	1,223.38
Ask	647.30 x 900	EPS (TTM)	0.52
Day's Range	610.20 - 642.75	Earnings Date	Jan 26, 2021 - Jan 31, 2021
52 Week Range	70.10 - 654.32	Forward Dividend & Yield	N/A (N/A)
Volume	51,113,780	Ex-Dividend Date	N/A
Avg. Volume	47,851,864	1y Target Est	389.79

Fair Value ⓘ 

XX.XX

Overvalued

[View details](#)

Related Research ⓘ 

 Analyst Report: Tesla, Inc.

 Market Update: CPB, ULTA, TSL...

[View more](#)

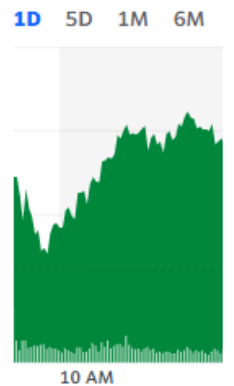


Chart Events ⓘ 

Bearish pattern detected

 **Fast Stochastic**

[View all chart patterns](#)

Tesla Valuation – Inputs

The “Input” section is where we place Tesla’s historical financial statements for the past 5 years. These will form the basis for our 10-year forecast model along with inputs from the “Drivers” sheet.

Input -->

	A	B	C	D	E	F	G
1	P&L Input						
2							
3	(in thousands)	2015 Act	2016 Act	2017 Act	2018 Act	2019 Act	
4	Automotive revenues	3,740,973	6,350,766	9,641,300	18,514,983	20,821,000	
5	Energy generation and storage	14,477	181,394	1,116,266	1,555,244	1,531,000	
6	Services and other	290,575	467,972	1,001,185	1,391,041	2,226,000	
7	Total revenues	4,046,025	7,000,132	11,758,751	21,461,268	24,578,000	
8	Automotive cost of revenues	(2,823,302)	(4,750,081)	(7,432,704)	(14,173,997)	(16,398,000)	
9	Energy generation and storage cost of revenues	(12,287)	(178,332)	(874,538)	(1,364,896)	(1,341,000)	
10	Services and other cost of revenues	(286,933)	(472,462)	(1,229,022)	(1,880,354)	(2,770,000)	
11	Gross profit	923,503	1,599,257	2,222,487	4,042,021	4,069,000	
12	Research and development	(717,900)	(834,408)	(1,378,073)	(1,460,370)	(1,343,000)	
13	Selling, general and administrative	(922,232)	(1,432,189)	(2,476,500)	(2,834,491)	(2,646,000)	
14	Restructuring and other				(135,233)	(149,000)	
15	EBIT	(716,629)	(667,340)	(1,632,086)	(388,073)	(69,000)	
16	Interest income	1,508	8,530	19,686	24,533	44,000	
17	Interest expense	(118,851)	(198,810)	(471,259)	(663,071)	(685,000)	
18	Other income (expense), net	(41,652)	111,272	(125,373)	21,866	45,000	
19	EBT	(875,624)	(746,348)	(2,209,032)	(1,004,745)	(665,000)	
20	Provision for income taxes	(13,039)	(26,698)	(31,546)	(57,837)	(110,000)	
21	Net loss incl. minority interests	(888,663)	(773,046)	(2,240,578)	(1,062,582)	(775,000)	
22	Minority interests	-	98,132	279,178	86,491	(87,000)	
23	Net loss	(888,663)	(674,914)	(1,961,400)	(976,091)	(862,000)	
24							
25	KPIs						
26	Revenue % automotive y-o-y		70%	52%	92%	12%	
27	Revenue % generation y-o-y		1153%	515%	39%	-2%	
28	Revenue % services y-o-y		61%	114%	39%	60%	

	A	B	C	D	E	F	G
1	Balance Sheet Input						
2							
3	(in thousands)	31 Dec 2015	31 Dec 2016	31 Dec 2017	31 Dec 2018	32 Dec 2019	
4	Cash and cash equivalents	1,196,908	3,393,216	3,367,914	3,685,618	6,268,000	
5	Restricted cash and marketable securities	22,628	105,519	155,323	192,551	246,000	
6	Accounts receivable, net	168,965	499,142	515,381	949,022	1,324,000	
7	Inventory	1,277,838	2,067,454	2,263,537	3,113,446	3,552,000	
8	Prepaid expenses and other current assets	115,667	194,465	268,365	365,671	713,000	
9	Total current assets	2,782,006	6,259,796	6,570,520	8,306,308	12,103,000	
10	Operating lease vehicles, net	1,791,403	3,134,080	4,116,604	2,089,758	2,447,000	
11	Solar energy systems, leased and to be leased, net	-	5,919,880	6,347,490	6,271,396	6,138,000	
12	Property, plant and equipment, net	3,403,334	5,982,957	10,027,522	11,330,077	10,396,000	
13	Operating lease right-of-use assets	-	-	-	-	1,218,000	
14	Intangible assets, net	12,816	376,145	361,502	282,492	339,000	
15	portion	-	506,302	516,889	489,707	591,000	
16	Restricted cash, net of current portion	31,522	268,165	441,722	398,219	269,000	
17	Other assets	46,858	216,751	273,123	571,657	808,000	
18	Total assets	8,067,939	22,664,076	28,655,372	29,739,614	34,309,000	
19	Accounts payable	916,148	1,860,341	2,390,250	3,404,451	3,771,000	
20	Accrued liabilities	422,798	1,210,028	1,731,366	2,094,253	2,905,000	
21	Deferred revenue	423,961	763,126	1,015,253	630,292	1,163,000	
22	Resale value guarantees	136,831	179,504	787,333	502,840	317,000	
23	Customer deposits	283,370	663,859	853,919	792,601	726,000	
24	Current portion of long-term debt and capital leases	627,927	984,211	796,549	2,567,699	1,785,000	
25	Current portion of solar bonds issued to related parties	-	165,936	100,000	-	-	
26	Total current liabilities	2,811,035	5,827,005	7,674,670	9,992,136	10,667,000	
27	Long-term debt and capital leases, net of current portion	2,021,093	5,860,049	9,415,700	9,403,672	11,634,000	
28	Other liabilities	2,152,107	5,438,936	6,330,414	4,586,166	4,541,000	



Workings

Tesla Valuation – Workings Section

The “Workings” sheet serves as a section divider in our model. It helps us distinguish the model’s inputs and its calculation sheets. This makes things easier and clearer for other people working with the model.

Workings -->

The “Workings” section of the model is further divided into “Income Statement Items” and “Balance Sheet Items”.

Balance Sheet -->

... **Workings -->** **Income statement items** Automotive Deliveries Deliveries development Deliveries comparables Revenue automotive

Income Statement items

The Income Statement section has 3 sub-sections:

- Automotive;
- Energy & Other; and
- Operating expenses.

The Balance Sheet section also has 3 sub-sections:

- PP&E;
- Working Capital; and
- Financing.

... **Workings -->** **Income Statement items** Automotive Deliveries Deliveries development De

... **Operating expenses** Opex comparables Opex **Balance Sheet -->** **PP&E -->** PP&E PP&E Comp

Deliveries

Vehicle	Tesla Deliveries								
	2015 Act	2016 Act	2017 Act	2018 Act	2019 Act	2020 Fcst	2021 Fcst	2022 Fcst	2023 Fcst
Tesla Model 3	-	-	1,764	146,055	300,815	481,304	529,434	582,378	628,968
Tesla Model S and X	50,580	76,230	101,312	98,865	66,385	69,040	71,802	74,674	77,661
Tesla Model Y	-	-	-	-	-	1,764	146,055	300,815	481,304
Tesla Roadster 2	-	-	-	-	-	-	-	500	1,000
Tesla Pickup	-	-	-	-	-	-	250	20,699	42,633
Tesla Semi	-	-	-	-	-	-	250	20,699	42,633
Total	50,580	76,230	103,076	244,920	367,200	552,108	747,791	999,766	1,274,198
y-o-y growth %									
Tesla Model 3	n.a.	n.a.	n.a.	8180%	106%	60%	10%	10%	8%
Tesla Model S and X	n.a.	51%	33%	-2%	-33%	4%	4%	4%	4%
Tesla Model Y	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	8180%	106%	60%
Tesla Roadster 2	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	100%
Tesla Pickup	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	8180%	106%
Tesla Semi	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	8180%	106%

We analyzed Tesla's historical deliveries of three main groups of products:

- Model S and Model X – established models with strong market performance;
- Model 3 – a new model with fast-growing sales, which went into production fairly recently;
- Other vehicles to be released in the future (Tesla Semi, Model Y etc.)

Tesla Valuation – Deliveries Forecast

$$I5 = H5 * (1 + I15)$$

Deliveries

Tesla Deliveries															
Vehicle	2015 Act	2016 Act	2017 Act	2018 Act	2019 Act	2020 Fcst	2021 Fcst	2022 Fcst	2023 Fcst	2024 Fcst	2025 Fcst	2026 Fcst	2027 Fcst	2028 Fcst	2029 Fcst
Tesla Model 3	-	-	1,764	146,055	300,815	481,304	529,434	582,378	628,968	679,286	706,457	734,715	764,104	794,668	826,455
Tesla Model S and X	50,580	76,230	101,312	98,865	66,385	89,040	71,802	74,674	77,661	80,768	83,998	87,358	90,852	94,487	98,266
Tesla Model Y	-	-	-	-	-	1,764	146,055	300,815	481,304	529,434	582,378	628,968	679,286	706,457	734,715
Tesla Roadster 2	-	-	-	-	-	-	-	500	1,000	1,500	1,650	1,815	1,960	2,117	2,202
Tesla Pickup	-	-	-	-	-	-	250	20,699	42,633	68,212	75,033	82,537	89,139	96,271	100,121
Tesla Semi	-	-	-	-	-	-	250	20,699	42,633	68,212	75,033	82,537	89,139	96,271	100,121
Total	50,580	76,230	103,076	244,920	367,200	552,108	747,791	999,766	1,274,198	1,427,411	1,524,549	1,617,929	1,714,481	1,790,270	1,861,881
y-o-y growth %															
Tesla Model 3	n.a.	n.a.	n.a.	8180%	106%	60%	10%	10%	8%	8%	4%	4%	4%	4%	4%
Tesla Model S and X	n.a.	51%	33%	-2%	-33%	4%	4%	4%	4%	4%	4%	4%	4%	4%	4%
Tesla Model Y	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	8180%	106%	60%	10%	10%	8%	8%	4%	4%
Tesla Roadster 2	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	100%	50%	10%	10%	8%	8%	4%
Tesla Pickup	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	8180%	106%	60%	10%	10%	8%	8%	4%
Tesla Semi	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	8180%	106%	60%	10%	10%	8%	8%	4%

Assumed growth after introduction:
 2 years 10%
 2 years 8%
 onwards 4%

Then, we used this data to project year-on-year growth rates:

- For established models we assumed 4% growth;
- For new models we assumed an introductory period of 3 years, largely mimicking Model 3's development (including forecasts for Q1 and Q2 in 2020);
- Then a gradual decline for a 4-year period until a "normal" growth rate of 4% is reached

Tesla Valuation – Deliveries Comparables

Deliveries comparables

Comparables

Vehicle Sales 2015-2019 - average	Average	Adj. Average
GM	9,132,000	9,600,340
Ford	5,966,667	6,607,000
Fiat Chrysler	4,500,000	4,740,000
BMW	2,366,667	2,460,000
Volkswagen	10,540,000	10,700,000
Jaguar	n.a.	n.a.
Porsche	n.a.	n.a.
Ferrari	n.a.	n.a.
Scania	n.a.	n.a.
MAN	n.a.	n.a.
Paccar	n.a.	n.a.
Average	n.a.	6,821,468

Roadsters and trucks, though part of the peer group in the reminder of the model for other purposes (such as profitability) have not been considered in this case.

Tesla Expected Deliveries	
2020	550,781
2024	1,382,049
2029	1,644,507

We took average vehicle deliveries over the past 5 years for several of Tesla's more established peers.
Clearly, our 10-year production forecast appear to be on the conservative side.

Tesla Valuation – Revenue Forecast

Revenue Automotive

H14 =CHOOSE(Drivers!\$C\$3,H22,H30,H38)

O5 =(\$R5*Deliveries!O5)*O14/1000000

	Tesla Revenue Automotive (in mln \$)											
Vehicle	2019 Act	2020 Fcst	2021 Fcst	2022 Fcst	2023 Fcst	2024 Fcst	2025 Fcst	2026 Fcst	2027 Fcst	2028 Fcst	2029 Fcst	Average Price (\$)
Tesla Model 3	14,706	26,066	28,672	31,540	34,063	36,788	38,259	39,790	41,381	43,037	44,758	54,157
Tesla Model S and	6,115	7,046	7,328	7,621	7,926	8,243	8,572	8,915	9,272	9,643	10,028	102,053
Tesla Model Y	-	108	8,909	18,350	29,360	32,295	35,525	38,367	41,436	43,094	44,818	61,000
Tesla Roadster 2	-	-	-	113	225	338	371	408	441	476	495	225,000
Tesla Pickup	-	-	15	1,209	2,490	3,984	4,382	4,820	5,206	5,622	5,847	58,400
Tesla Semi	-	-	44	3,622	7,461	11,937	13,131	14,444	15,599	16,847	17,521	175,000
Total	20,821	33,219	44,968	62,454	81,523	93,584	100,241	106,744	113,336	118,719	123,468	
Selected case												
Tesla Model 3		100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	
Tesla Model S and X		100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	
Tesla Model Y		100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	
Tesla Roadster 2		100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	
Tesla Pickup		100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	
Tesla Semi		100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	
Best case												
Tesla Model 3		102%	102%	102%	102%	102%	102%	102%	102%	102%	102%	
Tesla Model S and X		102%	102%	102%	102%	102%	102%	102%	102%	102%	102%	
Tesla Model Y		102%	102%	102%	102%	102%	102%	102%	102%	102%	102%	
Tesla Roads				102%	102%	102%	102%	102%	102%	102%	102%	
Tesla Pickup				102%	102%	102%	102%	102%	102%	102%	102%	
Tesla Semi				102%	102%	102%	102%	102%	102%	102%	102%	
Base case												
Tesla Model				100%	100%	100%	100%	100%	100%	100%	100%	
Tesla Model				100%	100%	100%	100%	100%	100%	100%	100%	
Tesla Model				100%	100%	100%	100%	100%	100%	100%	100%	
Tesla Roads				100%	100%	100%	100%	100%	100%	100%	100%	

Using the Choose function, we could further enhance the agility of the model by introducing Base, Best and Worst case scenarios, whereby prices increased or decreased (by 2%).

To project revenue, we obtained average prices through research and multiplied them by expected delivery

Using the Choose function, we could further enhance the agility of the model by introducing Base, Best and Worst case scenarios, whereby prices increased or decreased (by 2%).

To project revenue, we obtained average prices through research and multiplied them by expected delivery

Note: while the prices are given in USD at par, total revenues are given in millions USD as are all the financials going forward.

Tesla Valuation – Margins Automotive

GP %

Comparables 2019

Vehicle	Comparable companies	GP %	GP %	GP %
		Comp. 1	Comp. 2	Comp. 3
Tesla Model 3	GM, Ford, Fiat Chrysler	18.7%	14.3%	13.9%
Tesla Model S and X	BMW, Mercedes Cars, Volkswagen	15.5%	18.3%	17.2%
Tesla Model Y	GM, Ford, Fiat Chrysler	18.7%	14.3%	13.9%
Tesla Roadster 2	Jaguar, Porsche, Ferrari	36.0%	44.0%	52.3%
Tesla Pickup	GM, Ford, Fiat Chrysler	18.7%	14.3%	13.9%
Tesla Semi	Scania, MAN, Paccar	23.0%	23.0%	14.6%

Average

22.3%	Historic
22.3%	Historic
15.6%	Comparables
44.1%	Comparables
15.6%	Comparables
20.2%	Comparables

Historical

Vehicle	Historic Average GP%
Tesla Model 3	22%
Tesla Model S and X	22%
Tesla Model Y	n.a.
Tesla Roadster 2	n.a.
Tesla Pickup	n.a.
Tesla Semi	n.a.

In forecasting gross profit margins, we took historic margins for the models already in production. These are unique to Tesla's circumstances and hence a good predictor.

For the rest of the models we studied competitors' gross profit margins.

365 Careers

H14 =CHOOSE(Drivers!\$C\$3,H22,H30,H38)

O5 =O14*'Revenue automotive'!O5

[illegible]

Tesla Valuation – Cost of Sales

To calculate cost of sales:

Cost of sales = - (Revenues – Margins)

O5 = -('Revenue automotive'!O5-'GP automotive'!O5)

Cost of sales automotive

Vehicle	Tesla Cost of sales (in mln \$)														
	2015 Act	2016 Act	2017 Act	2018 Act	2019 Act	2020 Fcst	2021 Fcst	2022 Fcst	2023 Fcst	2024 Fcst	2025 Fcst	2026 Fcst	2027 Fcst	2028 Fcst	2029 Fcst
Tesla Model 3	n.a.	n.a.	n.a.	n.a.	n.a.	(20,242)	(22,266)	(24,492)	(26,452)	(28,568)	(29,711)	(30,899)	(32,135)	(33,420)	(34,757)
Tesla Model S and	n.a.	n.a.	n.a.	n.a.	n.a.	(5,471)	(5,690)	(5,918)	(6,155)	(6,401)	(6,657)	(6,923)	(7,200)	(7,488)	(7,788)
Tesla Model Y	-	-	-	-	-	(91)	(7,516)	(15,480)	(24,767)	(27,244)	(29,968)	(32,366)	(34,955)	(36,353)	(37,807)
Tesla Roadster 2	-	-	-	-	-	-	-	(63)	(126)	(189)	(208)	(228)	(247)	(266)	(277)
Tesla Pickup	-	-	-	-	-	-	(12)	(1,020)	(2,100)	(3,360)	(3,697)	(4,066)	(4,391)	(4,743)	(4,933)
Tesla Semi	-	-	-	-	-	-	(35)	(2,890)	(5,952)	(9,524)	(10,476)	(11,524)	(12,446)	(13,442)	(13,979)
Total	(2,823)	(4,750)	(7,433)	(14,174)	(16,398)	(25,804)	(35,519)	(49,862)	(65,552)	(75,286)	(80,716)	(86,006)	(91,374)	(95,712)	(99,541)

Tesla Valuation – Revenue Energy & Other

H10 =CHOOSE(Drivers!\$C\$3,H14,H17,H20)

O7 =N7*(1+O10)

Revenue Energy & Other

	Tesla Revenue Energy & Other (in mln \$)														
	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
(\$ in million)	Act	Act	Act	Act	Act	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst
Energy generation and storage	14	181	1,116	1,555	1,531	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Services and other	291	468	1,001	1,391	2,226	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Total Energy and Other	305	649	2,117	2,946	3,757	4,433	4,965	5,362	5,684	6,025	6,387	6,770	7,176	7,607	8,063
Selected case															
Total Energy and Other						18%	12%	8%	6%	6%	6%	6%	6%	6%	6%
Y-o-Y %															
Best case															
Total Energy and Other						20%	14%	10%	8%	8%	8%	8%	8%	8%	8%
Base case															
Total Energy and Other		113%	226%	39%	28%	18%	12%	8%	6%	6%	6%	6%	6%	6%	6%
Worst case															
Total Energy and Other						16%	10%	6%	4%	4%	4%	4%	4%	4%	4%

As revenues from energy, services and other sources was only a fraction of total revenue, we only forecast the total. We forecast a declining growth rate (as per historic trend).

Tesla Valuation – Margins Energy & Other

H11 =CHOOSE(Drivers!\$C\$3,H14,H18,H21)

O7 =O11*'Revenue Energy & Other'!O7

Gross Profit Energy & Other

	Tesla GP%Energy and Other (in mln \$)														
(\$ in million)	2015 Act	2016 Act	2017 Act	2018 Act	2019 Act	2020 Fcst	2021 Fcst	2022 Fcst	2023 Fcst	2024 Fcst	2025 Fcst	2025 Fcst	2027 Fcst	2028 Fcst	2029 Fcst
Energy generation and storage	2	3	242	190	190	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Services and other	4	(4)	(228)	(489)	(544)	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Total Energy and Other	6	(1)	14	(299)	(354)	-	-	-	-	-	-	-	-	-	-

Selected case															
Total Energy and Other	2%	0%	1%	-10%	-9%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
GP%															
Best case						2%	2%	2%	2%	2%	2%	2%	2%	2%	2%
Total Energy and Other															
Base case						0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Total Energy and Other															
Worst case						-2%	-2%	-2%	-2%	-2%	-2%	-2%	-2%	-2%	-2%
Total Energy and Other															

We looked at past GP margins to forecast gross profits for that revenue stream. Those were historically negative but our base assumption became that Tesla would achieve 0% margins over the forecast period.

Opex comparables

Comparables - 2019

Company	Average	Adj. Average
GM	15%	15%
Ford	13%	13%
Fiat Chrysler	9%	9%
BMW	10%	10%
Volkswagen	11%	11%
Jaguar	37%	
Porsche	31%	31%
Ferrari	28%	
Scania	15%	15%
MAN	15%	15%
Paccar	4%	4%
Average	17%	14%

We studied a select group of competitors' operating expenses as a share of revenue. We excluded certain outliers to balance our assumption.

D18 =AVERAGE(D7:D17)

Tesla Valuation – PP&E Comparables

PP&E comparables

Comparables - 2019

Company	CAPEX as a % of PP&E		CAPEX as a % of revenues	
	Average	Adj. Average	Average	Adj. Average
GM	-20%	-20%	-6%	-6%
Ford	-21%	-21%	-5%	-5%
Fiat Chrysler	-29%	-29%	-8%	-8%
BMW	-30%	-30%	-7%	-7%
Mercedes	-19%	-19%	-4%	-4%
Volkswagen	-22%	-22%	-6%	-6%
Jaguar	-24%	-24%	-7%	-7%
Ferrari	-33%	-33%	-9%	-9%
Paccar	-20%	-20%	-2%	
Average	-24%	-24%	-6%	-6%

We studied competitors' capital expenditures and used the results to project Capex for Tesla either as a share of beginning PP&E or as a share of revenue.

D16 =AVERAGE(D7:D15)

G16 =AVERAGE(G7:G15)

Tesla Valuation – Fixed Asset Roll Forward

365  Careers

B3 ="Selected case:"&CHOOSE(C3," as a % of PPE"," as a % of revenue")

Fixed asset roll forward

Selected case: as a % of PPE

1

H32 =CHOOSE(Drivers!\$C\$3,H36,H39,H42)
H33 =CHOOSE(Drivers!\$C\$3,H37,H40,H43)

H8 =CHOOSE(\$C\$3,H32*H7,H33*Opex!H5)

O7 =N10

O10 =SUM(O7:O9)

	Tesla PP&E (in mln \$)														
(\$ in million)	31Dec2015	31Dec2016	31Dec2017	31Dec2018	31Dec2019	31Dec2020	31Dec2021	31Dec2022	31Dec2023	31Dec2024	31Dec2025	31Dec2026	31Dec2027	31Dec2028	31Dec2029
	Act	Act	Act	Act	Act	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst
Beginning PP&E	738	3,403	5,983	10,028	11,330	10,396	12,912	16,037	19,919	24,740	30,727	38,164	47,401	58,873	73,122
Capex	3,088	3,527	5,681	3,204	1,220	2,516	3,125	3,881	4,821	5,988	7,437	9,237	11,472	14,249	17,698
D&A	(423)	(947)	(1,636)	(1,901)	(2,134)										
Ending PP&E	3,403	5,983	10,028	11,330	10,396	12,912	16,037	19,919	24,740	30,727	38,164	47,401	58,873	73,122	90,820

Selected case

Capex as a % of PP&E

Capex as a % of revenue

Capex %

Best case

Capex as a % of PP&E

Capex as a % of revenue

Base case

Capex as a % of PP&E

Capex as a % of revenue

Worst case

Capex as a % of PP&E

Capex as a % of revenue

Using Capex as % of beginning PP&E or revenues allows us to project expenditures.

We prepared our model in a way that allows us to choose whether to forecast Capex as % of PP&E or as % of revenue.

We can also choose a Best, Base or Worst-case scenario

24%	24%	24%	24%	24%	24%	24%	24%	24%	24%	24%	24%	24%	24%	24%	24%
6%	6%	6%	6%	6%	6%	6%	6%	6%	6%	6%	6%	6%	6%	6%	6%
19%	19%	19%	19%	19%	19%	19%	19%	19%	19%	19%	19%	19%	19%	19%	29%
3%	3%	3%	3%	3%	3%	3%	3%	3%	3%	3%	3%	3%	3%	3%	9%
24%	24%	24%	24%	24%	24%	24%	24%	24%	24%	24%	24%	24%	24%	24%	24%
6%	6%	6%	6%	6%	6%	6%	6%	6%	6%	6%	6%	6%	6%	6%	6%
29%	29%	29%	29%	29%	29%	29%	29%	29%	29%	29%	29%	29%	29%	29%	19%
9%	9%	9%	9%	9%	9%	9%	9%	9%	9%	9%	9%	9%	9%	9%	3%

H39 ='PP&E Comparables'!D16
H40 ='PP&E Comparables'!G16

Tesla Valuation – Fixed Asset Roll Forward

365  Careers

Fixed asset roll forward

Selected case: as a % of PPE

1

	Tesla PP&E (in mln \$)														
	31Dec2015	31Dec2016	31Dec2017	31Dec2018	31Dec2019	31Dec2020	31Dec2021	31Dec2022	31Dec2023	31Dec2024	31Dec2025	31Dec2026	31Dec2027	31Dec2028	31Dec2029
(\$ in million)	Act	Act	Act	Act	Act	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst
Beginning PP&E	738	3,403	5,983	10,028	11,330	10,396	11,621	12,861	14,090	15,275	16,378	17,351	18,140	18,680	18,900
Capex	3,088	3,527	5,681	3,204	1,220	2,516	2,813	3,113	3,410	3,697	3,964	4,199	4,390	4,521	4,574
D&A	(423)	(947)	(1,636)	(1,901)	(2,154)	(1,291)	(1,572)	(1,884)	(2,225)	(2,594)	(2,991)	(3,411)	(3,850)	(4,302)	(4,759)
Ending PP&E	3,403	5,983	10,028	11,330	10,396	11,621	12,861	14,090	15,275	16,378	17,351	18,140	18,680	18,900	18,714

Useful life Capex

10

Useful life historical assets

10

We assumed useful life of 10 years for both historic and newly acquired fixed assets

O10 =SUM(O7:O9)

O9 =O28

	Tesla D&A (in mln \$)														
	31Dec2015	31Dec2016	31Dec2017	31Dec2018	31Dec2019	31Dec2020	31Dec2021	31Dec2022	31Dec2023	31Dec2024	31Dec2025	31Dec2026	31Dec2027	31Dec2028	31Dec2029
(\$ in million)	Act	Act	Act	Act	Act	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst
						(1,040)	(1,040)	(1,040)	(1,040)	(1,040)	(1,040)	(1,040)	(1,040)	(1,040)	(1,040)
						(252)	(252)	(281)	(252)	(252)	(252)	(252)	(252)	(252)	(252)
									(281)	(281)	(281)	(281)	(281)	(281)	(281)
									(311)	(311)	(311)	(311)	(311)	(311)	(311)
									(341)	(341)	(341)	(341)	(341)	(341)	(341)
											(370)	(370)	(370)	(370)	(370)
											(396)	(396)	(396)	(396)	(396)
											(420)	(420)	(420)	(420)	(420)
												(439)	(439)	(439)	(439)
													(452)	(452)	(452)
														(457)	(457)
Total D&A	-	-	-	-	-	(1,291)	(1,572)	(1,884)	(2,225)	(2,594)	(2,991)	(3,411)	(3,850)	(4,302)	(4,759)

To complete our Fixed Asset Roll Forward, we added a depreciation schedule that would allow us to correctly project ending property, plant & equipment values for the entire forecast period.

H17 =-\$H\$7/\$C\$13
H18 =-\$H\$8/\$C\$13

I17 =-\$H\$7/\$C\$13
I18 =-\$H\$8/\$C\$13
I19 =-\$I\$8/\$C\$12

J17 =-\$H\$7/\$C\$13
J18 =-\$H\$8/\$C\$13
J19 =-\$I\$8/\$C\$12
J20 =-\$J\$8/\$C\$13

K17 =-\$H\$7/\$C\$13
K18 =-\$H\$8/\$C\$13
K19 =-\$I\$8/\$C\$12
K20 =-\$J\$8/\$C\$13
K21 =-\$K\$8/\$C\$12

O28 =SUM(O17:O27)

Tesla Valuation – Working Capital

Working capital

NB !!! To develop the Working Capital sheet, we first began to build Tesla's P&L, so we can have revenues and COGS coming from one place

O5 =O11*('P&L'!O5)/360
O6 =-O12*('P&L'!O\$6)/360
O7 =-O13*('P&L'!O\$6)/360
O8 =O5+O6-O7

Tesla Working capital (in mln \$)															
	31Dec2015	31Dec2016	31Dec2017	31Dec2018	31Dec2019	31Dec2020	31Dec2021	31Dec2022	31Dec2023	31Dec2024	31Dec2025	31Dec2026	31Dec2027	31Dec2028	31Dec2029
(\$ in million)	Act	Act	Act	Act	Act	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst
Trade receivables	169	499	515	949	1,324	1,920	2,546	3,458	4,447	5,080	5,438	5,789	6,146	6,442	6,708
Inventory	1,278	2,067	2,264	3,113	3,552	8,353	11,184	15,257	19,680	22,463	24,063	25,631	27,226	28,543	29,727
Trade payables	916	1,860	2,390	3,404	3,771	7,667	10,265	14,003	18,063	20,617	22,086	23,525	24,989	26,198	27,284
Net trade cycle	531	706	389	658	1,105	2,607	3,465	4,712	6,065	6,926	7,415	7,895	8,383	8,788	9,150

Selected case:															
DSO	15.0	25.7	15.8	15.9	19.4	18.4	18.4	18.4	18.4	18.4	18.4	18.4	18.4	18.4	18.4
DIO	147.3	137.8	85.4	64.3	62.3	99.5	99.5	99.5	99.5	99.5	99.5	99.5	99.5	99.5	99.5
DPO	105.6	124.0	90.2	70.4	66.2	91.3	91.3	91.3	91.3	91.3	91.3	91.3	91.3	91.3	91.3
Net trade cycle	56.7	39.5	11.0	9.9	15.5	26.5	26.5	26.5	26.5	26.5	26.5	26.5	26.5	26.5	26.5

We calculated historical DSO, DIO, and DPO (we have done it in the Balance Sheet Input sheet)

Then we used averages for the forecast period.

H11 =AVERAGE(\$C11:\$G11)
H12 =AVERAGE(\$C12:\$G12)
H13 =AVERAGE(\$C13:\$G13)
H14=H11+H12-H13

To calculate Trade receivables in the forecast period:
DSO * Revenues / 360
To calculate Inventory in the forecast period:
DIO * Cost of sales / 360
To calculate Trade payables in the forecast period:
DPO * Cost of sales / 360

Tesla Valuation – Financing

365  Careers

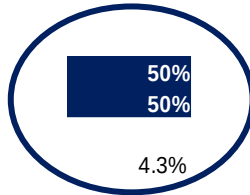
Financing

Cover negative cash flows with:

Debt

Equity

Bond Yield as of Sep. 20



When the company produces a negative cash flow, it will need additional financing during the forecast period.

We assumed such additional financing is covered with 50% debt and 50% equity. These percentages can be easily changed to test different hypothesis.

We also assumed that the cost of debt is equal to Tesla's 10-year bond yield-to-maturity (YTM) and remains constant throughout the entire forecast period.

	Tesla Financing (in mln \$)														
(\$ in million)	31Dec2015	31Dec2016	31Dec2017	31Dec2018	31Dec2019	31Dec2020	31Dec2021	31Dec2022	31Dec2023	31Dec2024	31Dec2025	31Dec2026	31Dec2027	31Dec2028	31Dec2029
	Act	Act	Act	Act	Act	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst
Long-term debt and capital leases, net of curre	2,021.1	5,860.0	9,415.7	9,403.7	11,634.0	11,634.0	11,634.0	11,634.0	11,634.0	11,634.0	11,634.0	11,634.0	11,634.0	11,634.0	11,634.0
Interest expense	(118.9)	(198.8)	(471.3)	(663.1)	(685.0)	(503.1)	(503.1)	(503.1)	(503.1)	(503.1)	(503.1)	(503.1)	(503.1)	(503.1)	(503.1)
Interest rate	-5.9%	-3.4%	-5.0%	-7.1%	-5.9%	4.3%	4.3%	4.3%	4.3%	4.3%	4.3%	4.3%	4.3%	4.3%	4.3%

	Tesla Cash Flow (in mln \$)														
	31Dec2015	31Dec2016	31Dec2017	31Dec2018	31Dec2019	31Dec2020	31Dec2021	31Dec2022	31Dec2023	31Dec2024	31Dec2025	31Dec2026	31Dec2027	31Dec2028	31Dec2029
	Act	Act	Act	Act	Act	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst
Unlevered Free Cash Flow						3,312.1	3,729.4	5,618.0	6,593.0	5,528.1	4,593.3	4,961.8	5,446.9	5,686.9	6,112.0
Debt						-	-	-	-	-	-	-	-	-	-
Equity						-	-	-	-	-	-	-	-	-	-

NB !!!

To develop the Financing sheet, we first needed to calculate Unlevered Free Cash Flow (UFCF), which necessitated the starting of both the Cash Flow Statement and the Balance Sheet

H18 ='Cash Flow'!H20
H19 =IF(H\$18<0,-H\$18*\$C4,0)
H20 =IF(H\$18<0,-H\$18*\$C5,0)

O11 =N11+N19
O12 =-N11*O13
O13 =\$C\$7

Tesla Valuation – WACC

WACC

From DRIVERS sheet

Risk-free rate	0.70%
Market risk premium	5.00%
Beta (as of 27 Sep 2018)	2.15
Tax rate	21%

Cost of equity
Current bond Yield as of 19th Sep

11.4%
4.3%

We calculated Tesla's cost of equity using the CAPM:
Risk-free + Beta * Market risk premium
And used its current YTM as cost of debt

C9 =C4+C5*C6
C10 =Drivers!C11

We calculated the portions of debt and equity financing

O18 =O15/(O\$15+O\$16)
O19 =O16/(O\$15+O\$16)

Tesla WACC calculation (in mln \$)

	31Dec2015	31Dec2016	31Dec2017	31Dec2018	31Dec2019	31Dec2020	31Dec2021	31Dec2022	31Dec2023	31Dec2024	31Dec2025	31Dec2026	31Dec2027	31Dec2028	31Dec2029
(\$ in million)	Act	Act	Act	Act	Act	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst
Debt	2649	6844	10212	11971	13419	13,419	13,419	13,419	13,419	13,419	13,419	13,419	13,419	13,419	13,419
Equity	1084	5538	5235	5758	7467	8,861	10,536	12,762	15,563	18,864	22,376	26,102	30,039	34,174	38,474
Debt / (Debt + Equity)	71%	55%	66%	68%	64%	60%	56%	51%	46%	42%	37%	34%	31%	28%	26%
Equity / (Debt + Equity)	29%	45%	34%	32%	36%	40%	44%	49%	54%	58%	63%	66%	69%	72%	74%
Cost of equity	11.4%	11.4%	11.4%	11.4%	11.4%	11.4%	11.4%	11.4%	11.4%	11.4%	11.4%	11.4%	11.4%	11.4%	11.4%
Cost of debt	4.3%	4.3%	4.3%	4.3%	4.3%	4.3%	4.3%	4.3%	4.3%	4.3%	4.3%	4.3%	4.3%	4.3%	4.3%
WACC	5.7%	7.0%	6.1%	6.0%	6.3%	6.6%	6.9%	7.3%	7.7%	8.1%	8.4%	8.7%	9.0%	9.2%	9.4%

NB !!!

To obtain WACC, we needed projections of debt and equity and so we could only prepare our WACC sheet after completion of the Balance Sheet

O24 =O18*O22*(1-SC\$7)+O19*O21

$WACC = E / (D+E) * \text{Cost of equity} + D / (D+E) * \text{Cost of debt} * (1-t)$

Output

A satellite view of Earth at night, showing city lights and cloud patterns over a curved horizon. The word "Output" is overlaid in white text on the left side.

Tesla Valuation – P&L

H5 ='Revenue automotive'!H11+'Revenue Energy & Other'!H7
H6 ='Cost of sales Energy & Other'!H7+'Cost of sales automotive'!H11

P&L

	Tesla P&L (in mln \$)														
(\$ in million)	2015 Act	2016 Act	2017 Act	2018 Act	2019 Act	2020 Fcst	2021 Fcst	2022 Fcst	2023 Fcst	2024 Fcst	2025 Fcst	2026 Fcst	2027 Fcst	2028 Fcst	2029 Fcst
Revenues	4,046	7,000	11,759	21,461	24,578	37,652	49,933	67,816	87,208	99,609	106,627	113,514	120,512	126,326	131,531
Cost of sales	(3,123)	(5,401)	(9,536)	(17,419)	(20,509)	(30,237)	(40,484)	(55,225)	(71,236)	(81,311)	(87,103)	(92,776)	(98,550)	(103,319)	(107,604)
Gross profit	924	1,599	2,222	4,042	4,069	7,415	9,449	12,591	15,971	18,298	19,524	20,738	21,962	23,007	23,927
Operating expenses	(1,640)	(2,267)	(3,855)	(4,430)	(4,138)	(5,147)	(6,826)	(9,271)	(11,922)	(13,617)	(14,576)	(15,518)	(16,475)	(17,269)	(17,981)
EBIT	(717)	(667)	(1,632)	(388)	(69)	2,268	2,623	3,320	4,050	4,681	4,948	5,220	5,487	5,737	5,946
Interest expenses/income	(159)	(79)	(577)	(617)	(596)	(503)	(503)	(503)	(503)	(503)	(503)	(503)	(503)	(503)	(503)
EBT	(876)	(746)	(2,209)	(1,005)	(665)	1,765	2,120	2,817	3,546	4,178	4,445	4,717	4,984	5,234	5,443
Taxes	(13)	(27)	(32)	(58)	(110)	(371)	(445)	(592)	(745)	(877)	(933)	(991)	(1,047)	(1,099)	(1,143)
Minority interest	-	98	279	86	(97)	-	-	-	-	-	-	-	-	-	-
Net Income	(889)	(675)	(1,961)	(976)	(862)	1,394	1,674	2,226	2,802	3,301	3,512	3,726	3,937	4,135	4,300

H12 =IF(H11>0,-H11*Drivers!\$C\$12,0)

We first filled in the P&L sheet up to EBIT, so that we have revenues and COGS in one place. This enabled us to project working capital accounts with ease.

We then used EBIT to project unlevered free cash flows (UFCF), which helped us project financing needs and, by extension, cost of financing

Finally, when we had derived debt financing costs (interest rate), we were able to complete the P&L by calculating tax expenses and net profit

Tesla Valuation – Balance Sheet

Balance Sheet

H6 = \$G\$6
H11 = \$G\$11
H12 = \$G\$12
H14 = \$G\$14

	Tesla BS (in mln \$)												
	31Dec2018	31Dec2019	31Dec2020	31Dec2021	31Dec2022	31Dec2023	31Dec2024	31Dec2025	31Dec2026	31Dec2027	31Dec2028	31Dec2029	
	Act	Act	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	
(\$ in million)													Assumption
Cash	3,686	6,268	9,183	12,515	17,735	23,931	29,062	33,257	37,822	42,871	48,161	53,875	Cash flow sheet
Restricted cash and securities	193	246	246	246	246	246	246	246	246	246	246	246	Stays flat
Trade receivables	949	1,324	1,920	2,546	3,458	4,447	5,080	5,438	5,789	6,146	6,442	6,708	Working capital sheet
Inventory	3,113	3,552	8,353	11,184	15,257	19,680	22,463	24,063	25,631	27,226	28,543	29,727	Working capital sheet
Prepaid expenses	366	713	867	1,150	1,561	2,008	2,293	2,455	2,614	2,775	2,909	3,028	Grows as a % of revenues
as a % of revenues	1.7%	2.9%	2.3%	2.3%	2.3%	2.3%	2.3%	2.3%	2.3%	2.3%	2.3%	2.3%	
Operating lease vehicles	2,090	3,665	3,665	3,665	3,665	3,665	3,665	3,665	3,665	3,665	3,665	3,665	Stays flat
Solar energy systems leases	6,271	6,138	6,138	6,138	6,138	6,138	6,138	6,138	6,138	6,138	6,138	6,138	Stays flat
PP&E	11,330	10,396	11,621	12,861	14,090	15,275	16,378	17,351	18,140	18,680	18,900	18,714	PP&E sheet
Intangible assets	282	339	339	339	339	339	339	339	339	339	339	339	Stays flat
Other assets	1,460	1,668	2,558	3,392	4,607	5,925	6,767	7,244	7,712	8,187	8,582	8,936	Grows as a % of revenues
as a % of revenues	6.8%	6.8%	6.8%	6.8%	6.8%	6.8%	6.8%	6.8%	6.8%	6.8%	6.8%	6.8%	
Total assets	29,740	34,309	44,890	54,036	67,097	81,654	92,431	100,197	108,095	116,273	123,925	131,377	

We filled in the Balance Sheet assets with items we have already forecasted (working capital and PP&E). We then modeled the rest as flat (not related to business activities) or as growing in sync with revenues (related to business activities).

Cash was initially left blank, and filled in last, upon completion of the Cash Flow Statement.

O9 =O10*('P&L'!O\$5)
O10 =AVERAGE(\$F\$10:\$G\$10)
O15 =O16*('P&L'!O\$5)
O16 =AVERAGE(\$F\$16:\$G\$16)

Tesla Valuation – Balance Sheet

Balance Sheet

H20 =H21*('P&L'!H\$5)	H24 =H25*('P&L'!H\$5)
H21 =AVERAGE(\$F\$21:\$G\$21)	H25 =AVERAGE(\$F\$25:\$G\$25)
H22 =H23*('P&L'!H\$5)	H26 =H27*('P&L'!H\$5)
H23 =AVERAGE(\$F\$23:\$G\$23)	H27 =AVERAGE(\$F\$27:\$G\$27)

	Tesla BS (in mln \$)												
	31Dec2018	31Dec2019	31Dec2020	31Dec2021	31Dec2022	31Dec2023	31Dec2024	31Dec2025	31Dec2026	31Dec2027	31Dec2028	31Dec2029	
(\$ in million)	Act	Act	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Fcst	Assumption
Accounts payable	3,404	3,771	7,067	10,265	14,003	18,063	20,617	22,086	23,525	24,989	26,198	27,284	Working capital sheet
Accrued liabilities	2,094	2,905	4,062	5,387	7,317	9,409	10,747	11,504	12,247	13,002	13,629	14,191	Grows as a % of revenues
as a % of revenues	9.8%	11.8%	10.8%	10.8%	10.8%	10.8%	10.8%	10.8%	10.8%	10.8%	10.8%	10.8%	
Deferred revenue	630	1,163	1,444	1,915	2,600	3,344	3,819	4,088	4,353	4,621	4,844	5,043	Grows as a % of revenues
as a % of revenues	2.9%	4.7%	3.8%	3.8%	3.8%	3.8%	3.8%	3.8%	3.8%	3.8%	3.8%	3.8%	
Resale value guarantees	503	317	684	907	1,232	1,584	1,809	1,937	2,062	2,189	2,295	2,389	Grows as a % of revenues
as a % of revenues	2.3%	1.3%	1.8%	1.8%	1.8%	1.8%	1.8%	1.8%	1.8%	1.8%	1.8%	1.8%	
Customer deposits	793	726	1,251	1,660	2,254	2,898	3,311	3,544	3,773	4,005	4,198	4,371	Grows as a % of revenues
as a % of revenues	3.7%	3.0%	3.3%	3.3%	3.3%	3.3%	3.3%	3.3%	3.3%	3.3%	3.3%	3.3%	
Long-term debt	11,971	13,419	13,419	13,419	13,419	13,419	13,419	13,419	13,419	13,419	13,419	13,419	Financing sheet
Other liabilities	4,586	4,541	7,501	9,948	13,511	17,374	19,845	21,243	22,615	24,009	25,167	26,204	Grows as a % of revenues
as a % of revenues	21.4%	18.5%	19.9%	19.9%	19.9%	19.9%	19.9%	19.9%	19.9%	19.9%	19.9%	19.9%	
Total liabilities	23,982	26,842	36,029	43,501	54,335	66,091	73,567	77,821	81,993	86,234	89,750	92,903	
Equity	5,758	7,467	8,861	10,536	12,762	15,563	18,864	22,376	26,102	30,039	34,174	38,474	
Total liabilities & equity	29,740	34,309	44,890	54,036	67,097	81,654	92,431	100,197	108,095	116,273	123,925	131,377	

Balance Sheet liabilities were filled in a similar fashion, except there were no flat items to forecast.

Long-term debt and Equity accounts were initially left blank, and filled in upon completion of the Finance sheet.

O29 =O30*('P&L'!H\$5)
O30 =AVERAGE(\$F\$30:\$G\$30)

Tesla Valuation – Cash Flow

365  Careers

Cash Flow

H6 =IF(H5*Drivers!\$C\$12>0,-H5*Drivers!\$C\$12,0)

H8=-'PP&E'!H9

	Tesla Cash Flow (in mln \$)														
(\$ in million)	2015 Act	2016 Act	2017 Act	2018 Act	2019 Act	2020 Fcst	2021 Fcst	2022 Fcst	2023 Fcst	2024 Fcst	2025 Fcst	2026 Fcst	2027 Fcst	2028 Fcst	2029 Fcst
EBIT		(667)	(1,632)	(388)	(69)	2,268	2,623	3,320	4,050	4,681	4,948	5,220	5,487	5,737	5,946
Operating taxes (21%)		(27)	(32)	(58)	(110)	(476)	(551)	(697)	(850)	(983)	(1,039)	(1,096)	(1,152)	(1,205)	(1,249)
NOPAT		(694)	(1,664)	(446)	(179)	1,792	2,072	2,623	3,199	3,698	3,909	4,124	4,335	4,532	4,697
Add-back D&A		947	1,636	1,901	2,154	1,291	1,572	1,884	2,225	2,594	2,991	3,411	3,850	4,302	4,759
Gross Cash Flow		253	(28)	1,455	1,975	3,083	3,644	4,507	5,424	6,293	6,900	7,535	8,185	8,834	9,457
Trade receivables		(330)	(16)	(434)	(375)	(596)	(626)	(912)	(989)	(632)	(358)	(351)	(357)	(296)	(265)
Inventory		(790)	(196)	(850)	(439)	(4,801)	(2,831)	(4,072)	(4,423)	(2,783)	(1,600)	(1,567)	(1,595)	(1,317)	(1,184)
Accounts payable		944	530	1,014	367	3,896	2,598	3,738	4,060	2,555	1,469	1,439	1,464	1,209	1,086
Investments in Working Capital		(176)	318	(269)	(447)	(1,502)	(859)	(1,247)	(1,352)	(861)	(489)	(480)	(488)	(405)	(363)
Capex		(3,527)	(5,681)	(3,204)	(1,220)	(2,516)	(2,813)	(3,113)	(3,410)	(3,697)	(3,964)	(4,199)	(4,390)	(4,521)	(4,574)
Other assets		(8,700)	(1,760)	1,820	(2,108)	(1,044)	(1,117)	(1,627)	(1,764)	(1,128)	(638)	(626)	(637)	(529)	(473)
Other liabilities		5,002	2,397	(2,212)	1,046	5,291	4,874	7,097	7,696	4,922	2,785	2,733	2,777	2,307	2,066
Unlevered Free Cash Flow		(7,147)	(4,753)	(2,410)	(754)	3,312	3,729	5,618	6,593	5,528	4,593	4,967	5,447	5,687	6,112

Cash flow calculation:

1. Calculate Operating taxes, to obtain NOPAT (Net Operating Profit After Tax);
2. Add back D&A (non-cash charge) to obtain Gross Cash Flow;
3. Calculate investments in working capital.

BEWARE! An increase in assets is a negative cash flow.
An increase in liabilities is a positive cash flow.

O11 =-('Balance Sheet'!O7-'Balance Sheet'!N7)
O12 =-('Balance Sheet'!O8-'Balance Sheet'!N8)
O13 ='Balance Sheet'!O19-'Balance Sheet'!N19

Tesla Valuation – Cash Flow

Cash Flow

H16 = -'PP&E'!H8

H17 = -('Balance Sheet'!H6-'Balance Sheet'!G6)-('Balance Sheet'!H9-'Balance Sheet'!G9)-('Balance Sheet'!H11-'Balance Sheet'!G11)-('Balance Sheet'!H12-'Balance Sheet'!G12)-('Balance Sheet'!H14-'Balance Sheet'!G14)-('Balance Sheet'!H15-'Balance Sheet'!G15)

	Tesla Cash Flow (in mln \$)														
(\$ in million)	2015 Act	2016 Act	2017 Act	2018 Act	2019 Act	2020 Fcst	2021 Fcst	2022 Fcst	2023 Fcst	2024 Fcst	2025 Fcst	2026 Fcst	2027 Fcst	2028 Fcst	2029 Fcst
EBIT		(667)	(1,632)	(388)	(69)	2,268	2,623	3,320	4,050	4,681	4,948	5,220	5,487	5,737	5,946
Operating taxes (21%)		(27)	(32)	(58)	(110)	(476)	(551)	(697)	(850)	(983)	(1,039)	(1,096)	(1,152)	(1,205)	(1,249)
NOPAT		(694)	(1,664)	(446)	(179)	1,792	2,072	2,623	3,199	3,698	3,909	4,124	4,335	4,532	4,697
Add-back D&A		947	1,636	1,901	2,154	1,291	1,572	1,884	2,225	2,594	2,991	3,411	3,850	4,302	4,759
Gross Cash Flow		253	(28)	1,455	1,975	3,083	3,644	4,507	5,424	6,293	6,900	7,535	8,185	8,834	9,457
Trade receivables		(330)	(16)	(434)	(375)	(596)	(626)	(912)	(989)	(632)	(358)	(351)	(357)	(296)	(265)
Inventory		(790)	(196)	(850)	(439)	(4,801)	(2,831)	(4,072)	(4,423)	(2,783)	(1,600)	(1,567)	(1,595)	(1,317)	(1,184)
Accounts payable		944	530	1,014	367	3,896	2,598	3,738	4,060	2,555	1,469	1,439	1,464	1,209	1,086
Investments in Working Capital		(176)	318	(269)	(447)	(1,502)	(859)	(1,247)	(1,352)	(861)	(489)	(480)	(488)	(405)	(363)
Capex		(3,527)	(5,681)	(3,204)	(1,220)	(2,516)	(2,813)	(3,113)	(3,410)	(3,697)	(3,964)	(4,199)	(4,390)	(4,521)	(4,574)
Other assets		(8,700)	(1,760)	1,820	(2,108)	(1,044)	(1,117)	(1,627)	(1,764)	(1,128)	(638)	(626)	(637)	(529)	(473)
Other liabilities		5,002	2,397	(2,212)	1,046	5,291	4,874	7,097	7,696	4,922	2,785	2,733	2,777	2,307	2,066
Unlevered Free Cash Flow		(7,147)	(4,753)	(2,410)	(754)	3,312	3,729	5,218	6,593	5,528	4,593	4,962	5,447	5,687	6,112

Cash flow calculation continued:

- Subtract capital investments;
- Subtract increase in other assets and add increase in other liabilities;
- Unlevered Free Cash Flow = Gross Cash Flow –(+) Investments in Working capital – Capex –(+) Changes in Other assets +(-) Changes in Other liabilities

H18 = ('Balance Sheet'!H20-'Balance Sheet'!G20)+('Balance Sheet'!H22-'Balance Sheet'!G22)+('Balance Sheet'!H24-'Balance Sheet'!G24)+('Balance Sheet'!H26-'Balance Sheet'!G26)+('Balance Sheet'!H29-'Balance Sheet'!G29)

O20 = O9+O14+O17+O18+O16

Tesla Valuation – Cash Flow

Cash Flow

	Tesla Cash Flow (in mln \$)														
(\$ in million)	2015 Act	2016 Act	2017 Act	2018 Act	2019 Act	2020 Fcst	2021 Fcst	2022 Fcst	2023 Fcst	2024 Fcst	2025 Fcst	2026 Fcst	2027 Fcst	2028 Fcst	2029 Fcst
EBIT		(667)	(1,632)	(388)	(69)	2,268	2,623	3,320	4,050	4,681	4,948	5,220	5,487	5,737	5,946
Operating taxes (21%)		(27)	(32)	(58)	(110)	(476)	(551)	(697)	(850)	(983)	(1,039)	(1,096)	(1,152)	(1,205)	(1,249)
NOPAT		(694)	(1,664)	(446)	(179)	1,792	2,072	2,623	3,199	3,698	3,909	4,124	4,335	4,532	4,697
Add-back D&A		947	1,636	1,901	2,154	1,291	1,572	1,884	2,225	2,594	2,991	3,411	3,850	4,302	4,759
Gross Cash Flow		253	(28)	1,455	1,975	3,083	3,644	4,507	5,424	6,293	6,900	7,535	8,185	8,834	9,457
Trade receivables		(330)	(16)	(434)	(375)	(596)	(626)	(912)	(989)	(632)	(358)	(351)	(357)	(296)	(265)
Inventory		(790)	(196)	(850)	(439)	(4,801)	(2,831)	(4,072)	(4,423)	(2,783)	(1,600)	(1,567)	(1,595)	(1,317)	(1,184)
Accounts payable		944	530	1,014	367	3,896	2,598	3,738	4,060	2,555	1,469	1,439	1,464	1,209	1,086
Investments in Working Capital		(176)	318	(269)	(447)	(1,502)	(859)	(1,247)	(1,352)	(861)	(489)	(480)	(488)	(405)	(363)
Capex		(3,527)	(5,681)	(3,204)	(1,220)	(2,516)	(2,813)	(3,113)	(3,410)	(3,697)	(3,964)	(4,199)	(4,390)	(4,521)	(4,574)
Other assets		(8,700)	(1,760)	1,820	(2,108)	(1,044)	(1,117)	(1,627)	(1,764)	(1,128)	(638)	(626)	(637)	(529)	(473)
Other liabilities		5,002	2,397	(2,212)	1,046	5,291	4,874	7,097	7,696	4,922	2,785	2,733	2,777	2,307	2,066
Unlevered Free Cash Flow		(7,147)	(4,753)	(2,410)	(754)	3,312	3,729	5,618	6,593	5,528	4,593	4,962	5,447	5,687	6,112
Interest expenses		(79)	(577)	(617)	(596)	(503)	(503)	(503)	(503)	(503)	(503)	(503)	(503)	(503)	(503)
Delta Financial liabilities		4,195	3,368	1,759	1,448	-	-	-	-	-	-	-	-	-	-
Delta equity/Other equity movements		5,129	1,658	1,499	2,571	-	-	-	-	-	-	-	-	-	-
Tax adjustment		-	-	-	-	106	106	106	106	106	106	106	106	106	106
Minority interest		98	279	86	(87)	-	-	-	-	-	-	-	-	-	-
Net Cash Flow		2,196	(25)	318	2,582	2,915	3,332	5,221	6,196	5,131	4,196	4,564	5,050	5,290	5,715

To reconcile Unlevered Free Cash Flow and Net Cash Flow, we must consider all items related to the financial structure of the firm, including:

- Interest expense, changes in debt and equity financing, minority interest,
- Adjust for the difference between taxes in the P&L and Operating taxes calculated here

O22 ='P&L'!O10
O23 ='Balance Sheet'!O28-'Balance Sheet'!N28
O24 ='Balance Sheet'!O32-'Balance Sheet'!N32-'P&L'!O14
O25 ='P&L'!O12-O6
O26 ='P&L'!O13

Tesla Valuation – DCF

We assumed a 2% perpetuity growth rate after the explicit forecast period.
Some industry practitioners use 3% alternatively.
This input is easily changed and tested.

DCF Valuation

g (Perpetuity growth)

2%

	2015 Act	2016 Act	2017 Act	2018 Act	2019 Act	1.0 2020 Fcst	2.0 2021 Fcst	3.0 2022 Fcst	4.0 2023 Fcst	5.0 2024 Fcst	6.0 2025 Fcst	7.0 2026 Fcst	8.0 2027 Fcst	9.0 2028 Fcst	10.0 2029 Fcst
Unlevered Free Cash Flow						3,312	3,729	5,618	6,593	5,528	4,593	4,962	5,447	5,687	6,112
Continuing value															84,587
Discount factor (WACC)						6.6%	6.9%	7.3%	7.7%	8.1%	8.4%	8.7%	9.0%	9.2%	9.4%
Present value of UFCF						3,107	3,261	4,544	4,895	3,743	2,825	2,764	2,740	2,579	2,496
Present value of CV															34,539

Perpetuity Growth Rate-Based Valuation

Total PV of Cash Flows	32,954
Continuing Value	84,587
PV of Continuing Value	34,539
Enterprise Value	67,493
+Cash	6,268
-Financial liabilities	13,419
Equity Value	60,342
Number of shares outstanding (as of 30 Sep 2020)	932
Price per share	64.7

F14 =SUM(H10:Q10)
F15 =Q7
F16 =Q11
F17 =F16+F14

H10 =H6/(1+H9)^H4
L10 =L6/(1+L9)^L4
O10 =O6/(1+O9)^O4

O7 =Q6*(1+\$C\$3)/(Q9-C3)
Q11 =Q7/(1+Q9)^Q4

Enterprise value calculation:

1. Calculate company Continuing Value (CV), which is what it will be worth after the forecast period;
2. Discount unlevered free cash flows to their present value (PV);
3. Discount company CV to present value (what it is worth today);
4. Sum discounted cash flows and PV of continuing value.

Tesla Valuation – DCF

DCF Valuation

g (Perpetuity growth)

2%

	2015 Act	2016 Act	2017 Act	2018 Act	2019 Act	1.0 2020 Fcst	2.0 2021 Fcst	3.0 2022 Fcst	4.0 2023 Fcst	5.0 2024 Fcst	6.0 2025 Fcst	7.0 2026 Fcst	8.0 2027 Fcst	9.0 2028 Fcst	10.0 2029 Fcst
(\$ in million)															
Unlevered Free Cash Flow						3,312	3,729	5,618	6,593	5,528	4,593	4,962	5,447	5,687	6,112
Continuing value															84,587
Discount factor (WACC)						6.6%	6.9%	7.3%	7.7%	8.1%	8.4%	8.7%	9.0%	9.2%	9.4%
Present value of UFCF						3,107	3,261	4,544	4,895	3,743	2,825	2,764	2,740	2,579	2,496
Present value of CV															34,539

Perpetuity Growth Rate-Based Valuation

Total PV of Cash Flows	32,954
Continuing Value	84,587
PV of Continuing Value	34,539
Enterprise Value	67,493
+Cash	6,268
-Financial liabilities	13,419
Equity Value	60,342
Number of shares outstanding (as of 30 Sep 2020)	932
Price per share	64.7

F17 =F16+F14
F18 ='Balance Sheet'!G5
F19 ='Balance Sheet'!G28
F20 =F17+F18-F19

F22 =F20/F21

Equity value and price per share calculation:

1. Calculate company Enterprise Value (EV), which is the PV of UFCF+CV;
2. Add present cash balance;
3. Subtract present long term liabilities;
4. This gives you intrinsic Equity Value at present.
5. Divide by the number of shares outstanding to obtain intrinsic share price.